

# NAWSHIN ALAM

*Environmental Oceanography — Ecosystem Modelling — Remote Sensing & GIS*

Dhaka, Bangladesh

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Personal Website | LinkedIn

## Education

### B.Sc. in Oceanography

Bangladesh Maritime University (BMU)

2020–2025

Bangladesh

- CGPA: 3.74/4.00.
- Position: 7th

## Research Experience

### Undergraduate Research Project: Evaluation of Ecosystem Health of the Saint Martin's Island Marine Protected Area of Bangladesh

#### Integrated Approach for Strategic Mangrove Restoration Plan

**Funded by:** United Nations Development Programme (UNDP)  
**Awarded in:** Youth4Climate 2025  
**Position:** Project Lead  
**Duration:** January 2026–December 2027 (Approx.)

### Understanding the Spatiotemporal Dynamics and Complexity of Urban Heat Islands Using Remote Sensing and Machine Learning

**Funded by:** Ministry of Science and Technology (MoST), Bangladesh  
**PD:** Maisha Farzana Methila  
**Position:** Research Associate  
**Duration:** March 2026–June 2026

### Assessing Ecosystem Health of a Deltaic Mangrove Island, Dubla Island, Sundarbans Mangrove Forest, through In Situ and Satellite Observation

**Funded by:** BMU  
**PI:** Dr. S. M. Mustafizur Rahman  
**Position:** Research Assistant  
**Duration:** July 2024–Present

### Water Quality Monitoring of the Major River Ecosystem of Dhaka City, Bangladesh

**Funded by:** BMU  
**PI:** Sheikh Fahim Faysal Sowrav  
**Position:** Research Assistant  
**Duration:** July 2024–July 2025

### Assessment and Zoning of Marine Protected Areas for Biodiversity Conservation in Bangladesh: A Comparative Study of Existing MPAs and Identification of Potential Sites

**Funded by:** MoST, Bangladesh  
**PI:** Dr. S. M. Mustafizur Rahman  
**Position:** Research Assistant  
**Duration:** July 2023–December 2024

## Publications

1. Hasan, M. A., Koly, F., Sowrav, F. F., Rahman, S. M. M., **Alam, N.**, Masud-Ul-Alam, M., Zuhaib Kabir, M., Sarder, S. R., & Ahmed, K. (2026). Assessing the dynamics of water quality and their implications on biochemical health of the river ecosystem: An RS-based study of the Shitalakshya River, Dhaka. *Bangladesh Maritime Journal*, 10(1), 2520–1840. <https://doi.org/10.70279/bmj-v2-1124>

## Technical Skills

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| <b>Programming Language</b> | Python, MATLAB, R, LaTeX, Ferret, SPSS                                                                                |
| <b>Software Skill</b>       | ArcGIS Pro, Google Earth Engine (GEE), MS Office, Canva, Adobe Illustrator, KoBoToolbox                               |
| <b>Instruments</b>          | Spectrophotometer, CTD, Algal Torch, Multiparameter, Terrameter, LiDAR, RTK GPS, Atomic Absorption Spectroscopy (AAS) |

## Awards & Fellowships

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- **National Science and Technology (NST) Fellowship**, MoST—Research on coastal productivity modelling.
- **Vice-Chancellor Scholarship**, Bangladesh Maritime University, 2025.

## Selected Training & Certifications

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- Workshop on **Coastal and Regional Ocean Community (CROCO) Model**, Bangladesh Maritime University, 2025.
- Training Course on **Numerical Model and Artificial Intelligence for Ocean Science (NMAI4OS)**, National Oceanographic and Maritime Institute.

## Conferences & Scientific Events

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- **25th International Conference of the North Indian Ocean Hydrographic Commission (NIOHC)**, Chattogram, Bangladesh, 10–12 May 2026.
- **Seabed Mapping: Enabling Ocean Action**, Dhaka, Bangladesh, 21 June 2025.
- **UN Decade of Ocean Science: Imperatives for Bangladesh 2021–2030**, Dhaka, Bangladesh, 17 November 2022.

## Leadership & Service

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|--------------------------------------------------------------|-----------------------------------------------|
| <b>Advisor</b><br><i>Maritime University Ocean Club</i>      | <b>May 2026–Present</b><br><i>BMU</i>         |
| <b>Communication Secretary</b><br><i>YouthMappers at BMU</i> | <b>Jul 2022–Jun 2023</b><br><i>Bangladesh</i> |

## Professional References

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### **Cdr. Nasrin Umme Nahar**

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Phone: +880 1678039526

### **Sheikh Fahim Faysal Sowrav**

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